

Supplementary

Pearson Correlation and Cosine Similarity

Since $\tilde{x} = (x_1, \dots, x_n)^\top$ and $\tilde{y} = (y_1, \dots, y_n)^\top$, define ℓ^2 -norm of vector $\tilde{x}$ is

$$\|\tilde{x}\|_{\ell^2} = \left(\sum_{i=1}^n |x_i|^2 \right)^{\frac{1}{2}}$$

The cosine similarity is

$$\cos(\theta) = \frac{\tilde{x}^\top \tilde{y}}{\|\tilde{x}\|_{\ell^2} \|\tilde{y}\|_{\ell^2}} = \frac{\sum_{i=1}^n x_i y_i}{\sqrt{(\sum_{i=1}^n x_i^2)(\sum_{i=1}^n y_i^2)}}$$

For centered data, the correlation coefficient can also be viewed as the cosine of the angle θ between the two observed vector in n-dimensional space. Let $\tilde{x}_c = (x_1 - \bar{x}, \dots, x_n - \bar{x})^\top$ and $\tilde{y}_c = (y_1 - \bar{y}, \dots, y_n - \bar{y})^\top$, the cosine similarity is

$$\cos(\theta) = \frac{\tilde{x}_c^\top \tilde{y}_c}{\|\tilde{x}_c\|_{\ell^2} \|\tilde{y}_c\|_{\ell^2}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{(\sum_{i=1}^n (x_i - \bar{x})^2)(\sum_{i=1}^n (y_i - \bar{y})^2)}} = r_{xy}$$

Quadratic form

Let $\tilde{\varepsilon} \sim N(0, \sigma^2 I_n)$ and let H is symmetric idempotent matrix of rank $p + 1$ then

$$\frac{\tilde{\varepsilon}^\top H \tilde{\varepsilon}}{\sigma^2} \sim \chi_{(r(H))}^2 = \chi_{(p+1)}^2$$

For two models ω and Ω , the difference in residual sum of square is

$$RSS_\omega - RSS_\Omega = \tilde{y}^\top (I_n - H_\omega - I_n + H_\Omega) \tilde{y} = \tilde{y}^\top (H_\Omega - H_\omega) \tilde{y} = \tilde{\varepsilon}^\top (H_\Omega - H_\omega) \tilde{\varepsilon}$$

and

$$RSS_\Omega = \tilde{y}^\top (I_n - H_\Omega) \tilde{y} = \tilde{\varepsilon}^\top (I_n - H_\Omega) \tilde{\varepsilon}$$

Hence

$$\frac{\tilde{\varepsilon}^\top (H_\Omega - H_\omega) \tilde{\varepsilon}}{\sigma^2} \sim \chi_{(r(H_\Omega) - r(H_\omega))}^2$$

and

$$\frac{\tilde{\varepsilon}^\top (I_n - H_\Omega) \tilde{\varepsilon}}{\sigma^2} \sim \chi_{(r(I_n) - r(H_\Omega))}^2$$

If $(I_n - H_\Omega)(H_\Omega - H_\omega) = 0$, these two quadratic forms are independent.

Therefore,

$$\frac{\frac{\tilde{\varepsilon}^\top (H_\Omega - H_\omega) \tilde{\varepsilon}}{\sigma^2} / (r(H_\Omega) - r(H_\omega))}{\frac{\tilde{\varepsilon}^\top (I_n - H_\Omega) \tilde{\varepsilon}}{\sigma^2} / (r(I_n) - r(H_\Omega))} \sim F_{r(H_\Omega) - r(H_\omega), r(I_n) - r(H_\Omega)}$$